10.1 Newton's Cradle

WORKSHEET

The question

The identical uniform smooth spherical marbles $A_1, A_2, \ldots, A_n$, where $n \geq 3$, each of mass m, lie in that order in a smooth straight trough, with each marble touching the next. The marble A_{n+1}, which is similar to A_n but has mass λm, is placed in the trough so that it touches A_n. Another marble A_0, identical to A_n, slides along the trough with speed u and hits A_1. It is given that kinetic energy is conserved throughout.

(i) Show that if $\lambda < 1$, there is a possible subsequent motion in which only A_n and A_{n+1} move (and A_0 is reduced to rest), but that if $\lambda > 1$, such a motion is not possible.

(ii) If $\lambda > 1$, show that a subsequent motion in which only A_{n-1}, A_n and A_{n+1} move is not possible.

(iii) If $\lambda > 1$, find a possible subsequent motion in which only two marbles move.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself A marble of mass m moving at speed u hits a stationary marble of mass M, and kinetic energy is conserved. What are the velocities afterwards? What if $M = m$?

STAGE 1 Only the last two move

Ask yourself Suppose only A_n (velocity p) and A_{n+1} (velocity q) move afterwards. Write down the conservation laws. Which marble's motion constrains the sign of p?

STAGE 2 Three marbles

Ask yourself Now suppose only A_{n-1}, A_n and A_{n+1} move, with velocities a, b, c. What inequalities must a, b, c satisfy, given which marbles are at rest and which are behind which?

Ask yourself Square the momentum equation and compare it with the energy equation. What is left?

Pause and try Notice the structure of the argument. Squaring the momentum equation produces cross terms like $2ab$, and the energy equation has none. The difference measures how "spread out" the motion is. Can you see why the same argument rules out *any* outcome in which only $A_k, \ldots, A_{n+1}$ move with non-negative velocities, when $\lambda > 1$?

STAGE 3 Two marbles that can move

Ask yourself If the marbles moving forwards must all be at the far end, and that fails, perhaps one marble moves *backwards*. Which marble is free to move backwards?

Try this next

1 **The same problem again.** (From STEP I, 2006.) Particles $A_1, \ldots, A_n$ lie at rest in a trough; A_{n-1} has mass m and A_n has mass λm with $\lambda > 1$. A particle A_0 of mass m hits A_1 with speed u. Show that it is impossible for exactly one particle to be

moving after all collisions, or for only A_{n-1} and A_n, or only A_{n-2}, A_{n-1}, A_n. Given that exactly two particles move, find their speeds.

2 **Following the collisions.** Model the impact in part (iii) as a sequence of separate two-marble collisions, starting with A_0 hitting A_1. Follow the sequence through and show that it ends in the motion found in part (iii).

3 **Two balls in.** In a cradle of five equal balls, two balls are released together and hit the other three with speed u. Show that "two balls leave with speed u" is consistent with both conservation laws, and find another outcome that is.

4 **Light end.** If $\lambda < 1$, follow the sequence of two-marble collisions. Do any marbles other than A_n and A_{n+1} end up moving?

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5. Conservation of momentum and energy, with many problems on collisions.
- **Frank Herrmann & Peter Schmälzle, "Simple explanation of a well-known collision experiment"**, *American Journal of Physics* 49 (1981). Why Newton's cradle works.
- **Dwight Neuenschwander, *Emmy Noether's Wonderful Theorem*** (Johns Hopkins University Press). An accessible account of the link between symmetry and conservation.

The full guided solution, and what the question leads to at university, is in Section 10.1:
https://beyondstep.keithmcnulty.org/ch10/newtons-cradle.html